EXPERIMENT NO. 4

AIM: To perform **INNER JOIN, LEFT JOIN, self-join, 3-way join, correlated subqueries, EXISTS, simulated INTERSECT and EXCEPT** queries on the Employee schema and compare execution plans using EXPLAIN.

SOURCE CODE:

```
USE companydb;
-- INNER JOIN
SELECT e.emp_name, d.dept_name
FROM Employee e INNER JOIN Department d
ON e.dept_id = d.dept_id;
```

Varun	Finance
Kriti	Finance
Yash	Finance
Tanya	Marketing
Dev	Marketing
Pooja	Marketing
Manav	Marketing
Simran	Marketing
Kabir	Marketing
Aditi	AI
Nakul	AI
Ira	AI
Aryan	AI
Diya	AI
Mohit	AI

```
-- LEFT JOIN
SELECT d.dept_name, e.emp_name
FROM Department d LEFT JOIN Employee e
ON d.dept_id = e.dept_id;
```

dept_name	emp_name
IT	Aarav
IT	Ananya
IT	Riya
IT	Karan
IT	Neha
IT	Rahul
HR	Ishita
HR	Aditya
HR	Priya
HR	Rohan
HR	Sneha
HR	Vivek
Finance	Meera
Finance	Arjun
Finance	Varun
Finance	Kriti
Finance	Yash
Marketing	Tanya
Marketing	Dev
Marketing	Pooja
Marketing	Manav
Marketing	Simran
Marketing	Kabir
AI	Aditi
AI	Nakul
AI	Ira
AI	Aryan
AI	Diya
AI	Mohit

-- SELF JOIN: Employee and Manager
SELECT e.emp_name AS Employee, m.emp_name AS Manager
FROM Employee e JOIN Employee m
ON e.manager_id = m.emp_id;

Arjun	Meera
Nisha	Meera
Varun	Meera
Kriti	Arjun
Yash	Arjun
Dev	Tanya
Pooja	Tanya
Manav	Tanya
Simran	Dev
Kabir	Dev
Nakul	Aditi
Ira	Aditi
Aryan	Aditi
Diya	Nakul
Mohit	Nakul

-- 3-WAY JOIN
SELECT e.emp_name, d.dept_name, p.project_name
FROM Employee e
JOIN Department d ON e.dept_id=d.dept_id
JOIN Project p ON d.dept_id=p.dept_id;

Rohan	HR	Recruitment
Sneha	HR	Recruitment
Vivek	HR	Recruitment
Meera	Finance	Payroll
Arjun	Finance	Payroll
Nisha	Finance	Payroll
Varun	Finance	Payroll
Kriti	Finance	Payroll
Yash	Finance	Payroll
Meera	Finance	FinanceAI
Arjun	Finance	FinanceAI
Nisha	Finance	FinanceAI
Varun	Finance	FinanceAI
Kriti	Finance	FinanceAI
Yash	Finance	FinanceAI

-- CORRELATED SUBQUERY
SELECT emp_name, salary, dept_id
FROM Employee e
WHERE salary >
(SELECT AVG(salary) FROM Employee
 WHERE dept_id=e.dept_id);

emp_name	salary	dept_id
Ananya	65000.00	1
Karan	72000.00	1
Rahul	58000.00	1
Ishita	62000.00	2
Rohan	68000.00	2
Vivek	56000.00	2
Meera	75000.00	3
Varun	65000.00	3
Tanya	70000.00	4
Manav	62000.00	4
Kabir	59000.00	4
Aditi	85000.00	5
Aryan	78000.00	5

```
-- EXISTS
SELECT d.dept_name
FROM Department d
WHERE EXISTS
(SELECT 1 FROM Employee e
 WHERE e.dept_id=d.dept_id);
```

dept_name
IT
HR
Finance
Marketing
AI

```
-- SIMULATED INTERSECT
SELECT emp_id FROM Employee WHERE dept_id=1
AND emp_id IN (SELECT emp_id FROM Employee WHERE salary>60000);
```

emp_id
2

```
-- SIMULATED EXCEPT
SELECT emp_id FROM Employee
WHERE dept_id=1
AND emp_id NOT IN (SELECT emp_id FROM Employee WHERE salary>60000);
```

emp_id
1
3
5
6

```
-- EXECUTION PLAN
EXPLAIN SELECT e.emp_name,d.dept_name
FROM Employee e JOIN Department d
ON e.dept_id=d.dept_id;
```

id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
1	SIMPLE	d	null	ALL	PRIMARY	null	null	null	5	100	null
1	SIMPLE	e	null	ref	dept_id	dept_id	5	db_454rzt3rg.d.dept_id	6	100	null